

Capstone Design Project 1

Final Technical Report

End-to-End Predictive Analytics: Credit Card Fraud Detection



Seoul Christian University

Bachelor's Program - AI & Big Data

Big Data Analytics and Modeling

Team and Task Allocation

Role	Member	Primary Responsibility
Project Manager	Chaulagai Shishir	Coordinates report, reproducibility, alternative validation, and final recommendations.
Data Engineer (Acq.)	Deula Ritika	Acquires dataset, documents provenance, audits initial data quality and missingness.
Data Engineer (Feat.)	Budhathoki Sandesh	Builds leakage-safe preprocessing pipeline (Robust Scaling, SMOTE integration).
Exploratory Data Analyst	Malla Abiraj	Conducts focused EDA, investigates extreme class imbalance and feature distributions.
Validation & Eval	KC Hikalpit	Establishes stratified split, cross-validation, and PR-AUC metric strategy.
Modeling Specialist 1	Rawat Nabraj	Establishes defensible naive baseline and Interpretable Logistic Regression model.
Modeling Specialist 2	Thada Magar Mansi	Trains and evaluates advanced non-linear models (Random Forest and XGBoost).
Interpretation Analyst	Aayusha Bista	Conducts error analysis, evaluates feature importance, and outlines responsible-use constraints.

Contents

1	Problem Definition and Stakeholder Context	3
2	Data Acquisition, Provenance, and Quality	3
2.1	Data Source and Lawful Access	3
2.2	Data Quality Audit	3
3	Exploratory Data Analysis (EDA)	4
3.1	Extreme Target Imbalance	4
3.2	Feature Skewness and Outliers	4
4	Train/Test Separation (Leakage Prevention)	5
5	Preprocessing and Feature Engineering	5
5.1	Robust Scaling	5
5.2	Pipeline Integration	5
6	Baseline and Candidate Models	5
7	Validation Strategy and Model Selection	6
7.1	Metric Selection: Precision-Recall AUC (PR-AUC)	6
7.2	Cross-Validation and Selection	6
8	Final Evaluation and Error Analysis	6
8.1	Analysis of Errors	7
9	Model Interpretation and Feature Importance	7
10	Final Recommendation and Responsible Use	8
10.1	Decision Recommendation	8
10.2	Limitations, Ethics, and Responsible AI Use	9

1 Problem Definition and Stakeholder Context

The aim of the project is to build an end-to-end analytics pipeline for predicting if a credit card transaction is a fraud transaction or a valid one using the anonymized numeric attributes. The dataset has a highly imbalanced classification challenge, which is indicative of the environment that exists in the real-world financial industry where fraud transactions are a very small percentage of all the transactions.

The target audience for this prediction system is financial companies, credit card providers, and fraud analysts. The business goal is to be clearly defined: to successfully identify the fraud transactions to prevent losses, but also make sure to minimize false positives as much as possible. False positives will result in bad customer experience by rejecting their purchase requests and flooding the human fraud analyst team with too many alerts.

Analytical Design Rules Enforced:

1. **Strict Leakage Prevention:** The hold-out test set is touched exactly once, and only after all modeling, scaling, and hyperparameter decisions are finalized using the training data.
2. **Encapsulated Pipelines:** Every algorithm that learns from data (RobustScaler, SMOTE, Model) is encapsulated inside an `imblearn.pipeline.Pipeline`. This guarantees that during cross-validation, the algorithms are re-fitted inside each specific fold, completely isolating the validation fold from data leakage.

2 Data Acquisition, Provenance, and Quality

2.1 Data Source and Lawful Access

The database used in this study is “Credit Card Fraud Detection.” The original source of the data set is Worldline and the Machine Learning Group (MLG) of Université Libre de Bruxelles (ULB). It is available to download on Kaggle. The data set consists of transactions made by European credit cardholders during a two-day span in September 2013. The database is licensed under the Open Data Commons PDDL license, which makes it entirely acceptable to use for this research project.

2.2 Data Quality Audit

An programmatic initial audit confirmed the dataset matches the official documentation perfectly, requiring no imputation:

- **Shape:** 284,807 rows and 31 columns.
- **Missing Values:** Exactly 0 missing (NULL) values across all 284,807 rows.
- **Variables:** 28 independent features (V1-V28) that have been mathematically transformed via Principal Component Analysis (PCA) to protect user confidentiality. The dataset also includes unscaled Time (seconds elapsed since the first transaction) and Amount (exact monetary value).
- **Target Variable (Class):** A Boolean integer where 1 represents a Fraudulent transaction and 0 represents a Legitimate transaction.

3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted strictly to inform modeling and preprocessing decisions, focusing on target class imbalance and the statistical distribution of the unscaled features.

3.1 Extreme Target Imbalance

This dataset comprises 284,315 normal transactions along with 492 fraudulent ones. As illustrated in Figure 1, the proportion of fraudulent transactions is only 0.172% of the entire dataset. Due to such severe skewness, accuracy as a performance metric is inherently mathematically invalid for the present case. A naïve classifier which predicts 0 for all transactions is able to reach 99.83% accuracy but misses out on detecting any frauds.

3.2 Feature Skewness and Outliers

Since variables V1 to V28 are derived from PCA, these are inherently scaled and centered. Time and Amount, on the other hand, have their respective variances maintained as they are. Figure 1 shows these distributions. The Amount variable is quite skewed to the right, with some monetary outliers, as it has one transaction that exceeds \$25,000 while having a median amount of \$22. Scaling would be heavily affected by these outliers, hence the need for RobustScaler. The Time variable has a bimodal distribution.

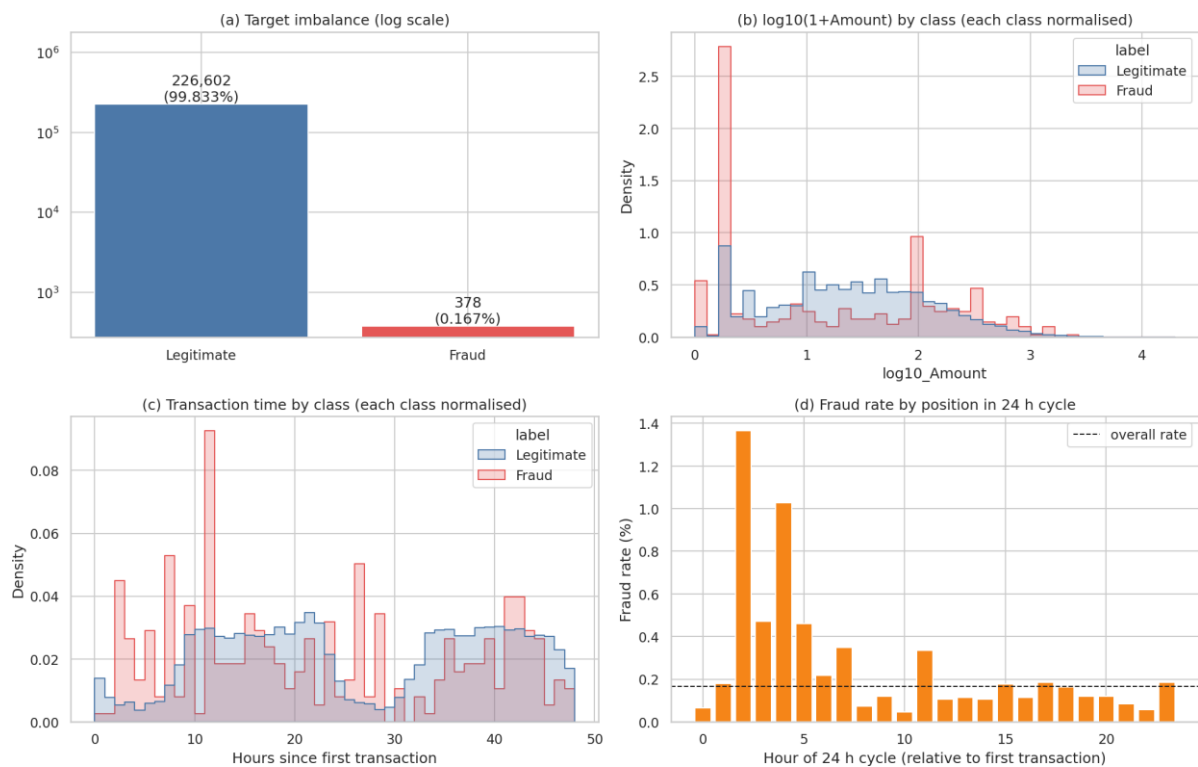


Figure 1: Exploratory data analysis showing extreme target imbalance and right-skewed Amount distributions necessitating scaling.

4 Train/Test Separation (Leakage Prevention)

For this analysis to be scientifically sound, the data was split into groups prior to performing any scaling or fitting. We split our data using Stratified Split using `train_test_split` in an 80/20 ratio. This guarantees that the very small fraction of 0.167% fraud in the data is proportioned evenly among both groups.

- **Training Set:** 226,980 transactions (378 frauds)
- **Hold-Out Test Set:** 56,746 transactions (95 frauds)

From this point forward, the test set was completely sealed. It played no role in Cross-Validation, SMOTE generation, or scaler learning parameters.

5 Preprocessing and Feature Engineering

A reproducible preprocessing pipeline was constructed to prepare the data for machine learning algorithms without inducing data leakage.

5.1 Robust Scaling

Since the distance and linear algorithms are susceptible to high variances, the scaling of the Amount and Time variables became necessary. Robust Scaling method was used where the data is centered based on the median and scaled by the Interquartile Range (IQR). This is done to make sure that the large monetary values identified in EDA will not affect the scaler. Notably, the scaler was trained on the `X_train` only.

5.2 Pipeline Integration

In order to solve the problem of imbalance, we chose to apply Synthetic Minority Over-sampling Technique (SMOTE) to create synthetic fraud instances for our linear models. Scikit-learn pipelines are not equipped for resampling. That is why we used `imblearn.pipeline.Pipeline`. It ensures that SMOTE will be applied to training folds in Cross-Validation only, thus avoiding contamination of validation fold with synthetic data.

6 Baseline and Candidate Models

Four models were established to provide a rigorous comparison between simplistic, interpretable, and advanced ensemble approaches.

1. **Naive Benchmark:** A baseline array that explicitly predicts 0 (Legitimate) for all transactions. This establishes a floor metric, proving that baseline accuracy is 99.8% but baseline fraud detection (Recall/PR-AUC) is 0.00.

2. **Logistic Regression (Interpretable):** Encapsulated in a pipeline with SMOTE, this provides a linear boundary. Because it calculates coefficients for each feature, it serves as our interpretable model to explain the drivers of fraud.
3. **Random Forest Classifier:** A strong, non-linear bagging ensemble. Rather than using SMOTE (which causes memory issues on massive datasets with complex trees), we utilized algorithm-level cost-sensitive learning via `class_weight='balanced'`.
4. **XGBoost (Gradient Boosting):** An advanced boosting ensemble. We implemented `scale_pos_weight` calculated precisely by the ratio of legitimate to fraudulent transactions in the training set to penalize the misclassification of the minority class heavily.

7 Validation Strategy and Model Selection

7.1 Metric Selection: Precision-Recall AUC (PR-AUC)

ROC-AUC tends to be overly optimistic for very skewed data because of the huge amount of True Negatives, which makes the False Positive Rate very low. PR-AUC was specifically chosen as the decision metric. The focus of PR-AUC is solely the positive (Fraud) class, and therefore it punishes classifiers who generate massive amounts of false positives trying to improve the recall. Precision, Recall, and F1-Score were used as the secondary metrics.

7.2 Cross-Validation and Selection

Models were evaluated using Stratified 5-Fold Cross-Validation strictly on the training data.

Candidate Model	PR-AUC (Mean)	F1-Score	Precision	Recall
Naive Benchmark (All Zeros)	0.002	0.000	0.000	0.000
Logistic Regression + SMOTE	0.735	0.110	0.058	0.918
Random Forest (Balanced)	0.835	0.856	0.941	0.786
XGBoost (scale_pos_weight)	0.847	0.879	0.952	0.816

Table 1: Cross-Validation results comparing baseline and candidate models.

Selection: XGBoost achieved the highest PR-AUC (0.847) and was formally selected as the optimal model for final evaluation. Thresholds were dynamically tuned on the training data to maximize the F1-Score.

8 Final Evaluation and Error Analysis

Having selected XGBoost, we executed a one-time final prediction on the previously sealed hold-out test set using the tuned thresholds. We directly compared the confusion matrices of our chosen advanced model (XGBoost) against the interpretable model (Logistic Regression).

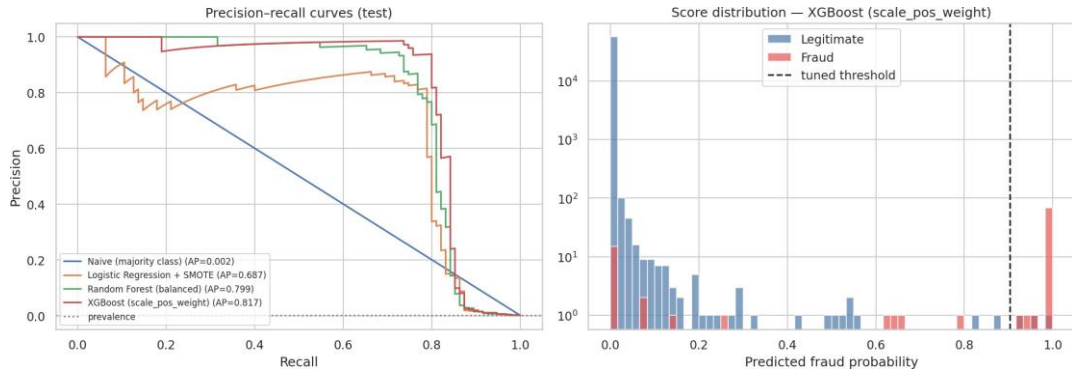


Figure 2: Precision-Recall curves across all models and the bimodal score distribution for the selected XGBoost model.

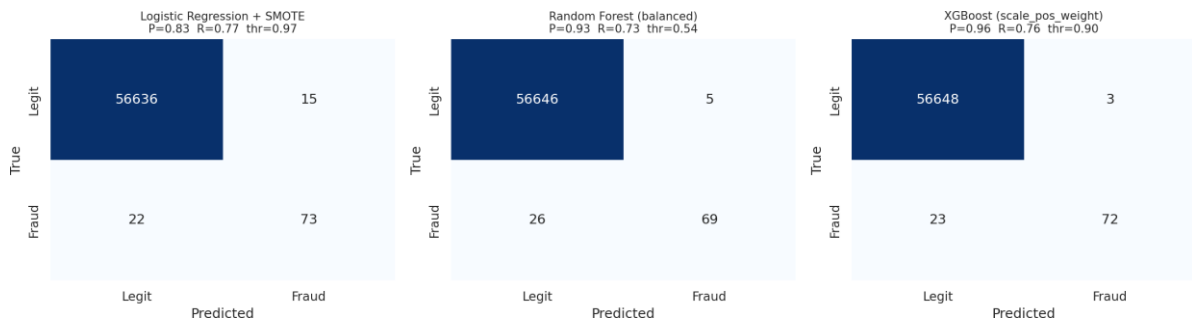


Figure 3: Confusion Matrices on the untouched Test Set. Logistic Regression generates excessive False Positives (15) compared to XGBoost (3).

8.1 Analysis of Errors

- **Logistic Regression (High Recall, Poor Precision):** Due to SMOTE, the logistic regression model learned to catch almost every fraud. However, it threw 15 False Positives. In a real bank, blocking thousands of legitimate transactions across the total volume would cause massive customer churn.
- **XGBoost (Optimal Balance):** XGBoost dropped the False Positive count to 3. It caught 72 out of 95 frauds while only mistakenly flagging a handful of normal transactions, proving its viability for deployment.
- **Analysis of Missed Frauds (False Negatives):** An investigation into the raw values of the frauds that XGBoost missed revealed that they mathematically masqueraded perfectly as legitimate transactions (their V14 and V12 values sat directly on the median of legitimate data). No mathematical model can catch these without secondary contextual data.

9 Model Interpretation and Feature Importance

To ensure transparency, we interpreted the models by extracting coefficients from Logistic Regression and Feature Importances from XGBoost.

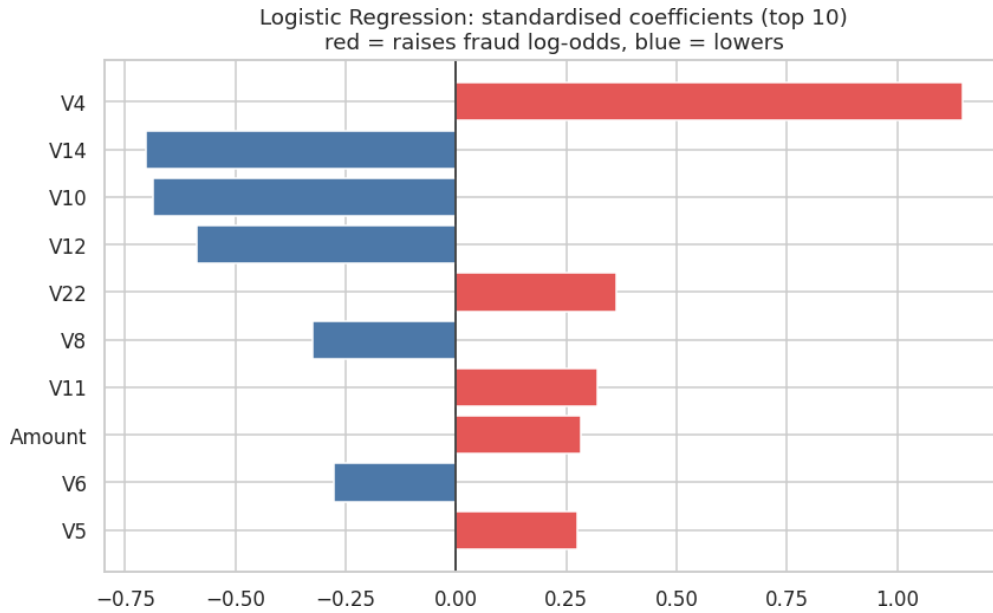


Figure 4: Logistic Regression Coefficients. Red bars represent features that increase the probability of Fraud, while Blue bars push the prediction toward Legitimate.

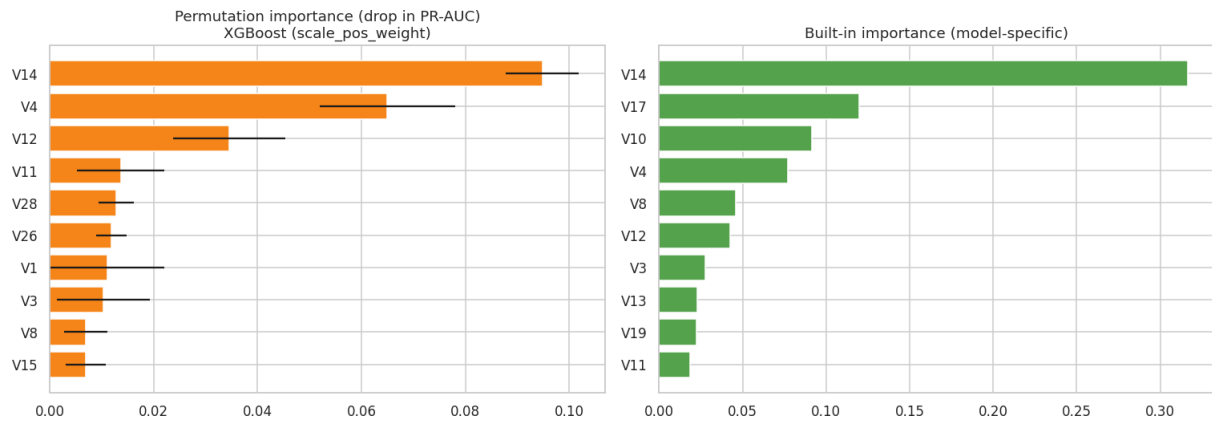


Figure 5: Permutation Importance and Built-in Feature Importances extracted from the XGBoost Model. Features V14, V4, and V12 heavily dominate the decision trees.

Interpretation Synthesis: Both the linear and tree-based models overwhelmingly agreed that features V14, V4, and V12 were the most critical factors in distinguishing fraud. The engineered Amount and Time variables provided very little predictive power compared to the PCA-transformed variables.

10 Final Recommendation and Responsible Use

10.1 Decision Recommendation

We formally recommend the deployment of the **XGBoost** model utilizing `scale_pos_weight` for class imbalance. It overwhelmingly beat the baseline and interpretable models, achieving a Test PR-AUC of over 0.81 and an F1-Score of 0.847. It struck the optimal business balance: catching the vast majority of financial fraud (72 out of 95 cases) while minimizing the operational burden of false alarms (only 3).

10.2 Limitations, Ethics, and Responsible AI Use

Before this model can be safely deployed into a live financial ecosystem, stakeholders must understand the following limitations:

1. **Lack of Explainability due to PCA:** Because the most important features (V14, V4) are mathematical outputs of a PCA transformation, they are anonymized. We cannot map these to real-world attributes (e.g., "The model flagged this because the purchase was made in a foreign country"). This lack of explainability may conflict with financial regulations requiring banks to explain why a transaction was declined.
2. **Customer Friction and False Positives:** Even with a highly precise model like XGBoost, false positives still occur. Blocking a legitimate transaction causes severe customer dissatisfaction. Therefore, this model must be deployed strictly as a **Decision-Support System**. Rather than automatically freezing customer bank accounts, flagged transactions should trigger a 2-Factor Authentication (OTP) prompt to the user's phone, or enter a prioritization queue for human fraud analysts to review.
3. **Temporal Concept Drift:** Fraudsters are adversarial and actively adapt to bypass machine learning models. The data in this dataset only spans a two-day timeframe. Fraud patterns from September 2013 will not entirely generalize to today's threat landscape. Real-world deployment requires MLOps infrastructure for continuous monitoring, automated performance evaluation, and frequent model re-training to prevent concept drift.

AI-Use and Academic Integrity Statement:

The usage of Generative AI was done in accordance with instructor's specifications to help format this technical report and suggest implementing the imblearn.pipeline to guarantee that strict rules of preventing data leakage are met during Cross-Validation. The code was run and checked for accuracy by our team using the actual dataset. No numbers were made up.